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CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Ho Chi Minh, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
20/10/26	A	Ngô Quang Khôi	Initial Software Test Documentation: scope, strategy, plan, automated and manual test cases, and the test report of the 20/10/2026 execution.
31/10/26	M	Mai Quốc Anh	Matched automated rows to the 31/10/2026 checkout: Passed where the assertion ran, Skipped for the seven unconfigured integration tests, and Not mapped where no executed assertion exists.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

1.1 System boundaries

The system under test consists of three separate repositories. The Arden monorepo contains browser and desktop clients, shared React UI and domain packages, Core API, migrations, worker and release tooling. Arden-AI contains FastAPI endpoints, document compilation, Chroma integration, answering and indexing contracts. Arden-Knowledge contains engineering/product/operations documentation and, in the pinned snapshot, no executable test modules.

Area	Implemented or tested scope	Outside this report's verified end-to-end evidence
Authentication	Email registration, password recovery, Supabase session adapter, Google PKCE, desktop API request policy	Real provider availability, full browser callback and deliverability UAT
Organizations	Organization creation, department membership, invitations, admin operations, reviewer grants	Multi-tenant load and external-identity administration
Private workspace	Private note state, version conflicts, PDF bytes, owner access enforcement, content references	Cross-device offline synchronization
Shared knowledge	Shared draft, immutable submission, classification, audience, review decision, work queue	Unimplemented multi-stage policy or workflow assumptions
Publication indexing	Manifest contracts, leased indexing worker, snapshot hashes, PDF extraction and chunking	Large-scale indexing throughput
Ask Arden	Answer request contract, service auth, core generation allowlist, citation grounding and revocation recheck	Quantitative model answer-quality benchmark
Release	Checksum-aware migrations, protocol identity, rollback, retention, CI schema isolation	An executed production cutover in inspected PR CI

1.2 Priority and business impact

A critical case is one where incorrect behavior could leak restricted information, grant unauthorized membership or review permissions, corrupt or mispublish governed content, or make a production upgrade unsafe. High-priority cases protect usability, recovery and deterministic content behavior. Priorities in the companion workbook are a proposed QA triage classification, not a previously approved Arden severity policy.

Priority	Typical risk	Examples from this test plan
Critical	Confidentiality, authorization, publication integrity, release integrity	Cross-actor note denial, reviewer-grant validation, citation recheck, tampered release
High	Recoverability, user state, content interpretation or reliable delivery	Conflict recovery, email configuration safety, chunking stability, document navigation

1.3 Exclusions and assumptions

- The code contains a desktop package and a main-process test; that does not by itself demonstrate a packaged Electron installer has passed end-to-end acceptance.
- No browser automation harness was identified as a successful full workflow smoke test in the inspected source inventory. UI component tests are treated as model/component evidence.
- CI checks use disposable environments. They should never be re-pointed at production storage, identity tenants or retained document data.
- Synthetic fixtures from tests are valid for automated verification, but they are not representative load, penetration or user-acceptance measurements.

1.4 Functional and non-functional test scope

Requirement category	Acceptance concern	Current evidence / additional work
Functional – account and organization	Register, authenticate, invite, assign membership and reject stale or cross-organization access	API/component tests exist; complete staged actor journeys pending
Functional – knowledge lifecycle	Save private versions, copy shared draft, freeze submission, review, publish, index and retrieve	Policy, API and worker tests exist; full cross-service journey pending
Security and privacy	Deny unauthorized reads, policy bypass, forged service requests, invalid citations and tampered releases	Static negative tests and disposable PostgreSQL job; independent security assessment not supplied
Reliability and recovery	Recover after version conflict, unavailable database/AI service, failed index or interrupted release	Automated recovery paths exist; staged incident drill pending
Performance and capacity	Define p95/p99 response, concurrent member, indexing throughput and PDF-size targets before load testing	No approved performance thresholds or measured load report provided
Usability and accessibility	Check keyboard navigation, visible errors, status announcements, contrast and screen-reader flow	No completed accessibility audit or device matrix supplied
Compatibility and portability	Run Node and release suites on Windows/Linux; verify packaged Electron and browser targets	CI Windows/Linux jobs succeeded; packaged desktop and browser matrix not verified

A non-functional concern without an agreed threshold is documented as a test-design gap, not marked Passed or Failed. The exit decision must name the approved criterion, the measured sample, environment and run artifact.

2. Test Strategy

Arden is tested bottom-up. Fast automated unit and API tests run on every change and in CI on Ubuntu and Windows; database integration tests verify the real schema, migrations and row-level security; system and acceptance tests exercise the deployed product through its clients. Authorization and privacy rules receive negative tests at every level because they are the core promise of the product.

2.1 Testing Types

The testing types applied in the project are described below.

Testing type	Objective	Technique	Completion criteria
Functional testing	Verify that each	Unit tests of policies and view	All functional test

	implemented use case and business rule behaves as specified.	models; API route tests with Fastify inject; equivalence partitioning, boundary values (5 MB PDF limit, 7-day invitation expiry) and state-transition tests (submission and invitation states).	cases pass; every implemented use case has at least one automated test.
Authorization and privacy testing	Verify owner-only private notes, audience-based reading, organization isolation, department boundary, reviewer-grant scope, self-review denial and administrator separation.	Negative tests with forged identifiers and cross-organization guesses; denial matrix tests in the policy layer; repository tests under the non-bypass arden_runtime role with row-level security.	All denial cases pass; responses never reveal hidden titles, counts or existence.
Data integrity and concurrency testing	Verify optimistic versions, idempotent create/submit/decide, immutable snapshot and hash binding, and atomic approval with publication.	Stale If-Match versions, replayed Idempotency-Keys, mismatched snapshot hashes, simulated storage or commit failures.	No lost update, duplicate record or partially published version is produced.
Input validation and error handling	Verify that malformed or oversized input is rejected safely.	JSON-schema violations, unknown fields, NUL bytes, oversized bodies, malformed UUIDs and file types.	Invalid input returns a 4xx safe error without internal details.
Integration testing	Verify that components work together: API with database migrations and RLS, UI transport with the desktop bridge.	Apply migrations 0001-0017 to a clean PostgreSQL; run the repository integration suite; client transport tests against the desktop bridge contract.	Migrations apply cleanly and the integration suite passes.
Regression testing	Detect regressions on every push and pull request.	GitHub Actions pipeline on Ubuntu and Windows: install, typecheck, all test suites, build.	Pipeline is green before merge.
UI and usability testing	Verify screen flows, layout and accessibility of the web and desktop clients.	Manual checklist in Chromium at 1440, 1024, 720, 390 and 320 px widths; keyboard navigation; desktop renderer checks.	No blocking layout or navigation issue in the checklist.
Compatibility testing	Verify supported platforms.	Web on current Chromium-based browsers and Firefox; Windows desktop (Electron); CI on Ubuntu and Windows.	Main flows work on every supported platform.
Performance testing (planned)	Measure API latency and concurrent-user behaviour on the hosted deployment.	Load scripts against a hosted slot with the demo dataset in W12.	Thresholds to be agreed before W12 (NFR-PERF-05).
Deployment and infrastructure testing	Verify hosted development protocol, maintenance fences and remote tooling.	Python unittest for hosted sync/fence scripts; Node test runner for the remote development CLI.	All infrastructure tests pass.

2.2 Test Levels

The table shows at which levels each testing type is performed.

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
Functional testing	X	X	X	X
Authorization and privacy testing	X	X	X	X
Data integrity and concurrency testing	X	X	X	
Input validation and error handling	X	X	X	
Integration testing		X		
Regression testing	X	X	X	
UI and usability testing			X	X
Compatibility testing			X	X
Performance testing (planned)			X	
Deployment and infrastructure testing	X	X	X	

2.3 Supporting Tools

The following tools support testing in the project.

Purpose	Tool	Vendor/In-house	Version
Unit and component runner (TypeScript)	Vitest	Open source	Monorepo lockfile-controlled
API route tests (in-process)	Fastify inject	Open source	Fastify / pinned lockfile
Desktop and release tests	Node.js test runner (node:test)	OpenJS Foundation	Node 24 CI baseline
AI service tests	pytest with uv	Open source	Python >=3.11,<3.14; run unverified
Static type checking	TypeScript compiler (tsc)	Microsoft	Pinned monorepo lockfile
Workspace scripts	pnpm	Open source	11.19.0 workflow baseline
Continuous integration	GitHub Actions	GitHub	CI #268; Ubuntu and Windows
Disposable database integration	PostgreSQL + pgvector	Open source	PostgreSQL 17 CI fixture
Schema migration safety	migrate.ts and operator checks	In-house	Pinned project source
Synthetic actors and authorization	Local test-mode guard / fixtures	In-house	Loopback and disposable DB only
Manual system and acceptance	Staging browser / Windows desktop	In-house	Runs pending; record build and browser

Defect and change tracking	GitHub Issues and pull requests	GitHub	Link exact build and evidence

3. Test Plan

3.1 Human Resources

The project members involved in testing and their responsibilities are listed below.

Worker/Doer	Role	Specific Responsibilities/Comments
Ngô Quang Khôi	Test Lead	Owns the test plan and this report; maintains the CI pipeline; reviews test evidence for every pull request; consolidates defects and the final test report.
Đặng Trường Hưng	Tester / Developer	Writes and runs unit, API and system tests for Mainflow 1 (sign-in, organizations, departments, invitations, audience groups).
Mai Quốc Anh	Tester / Developer	Writes and runs unit, API and system tests for Mainflow 2 (private notes, shared drafts, submission, review and publication).
Lâm Gia Kiệt	Tester / Environment owner	Maintains the local test mode and isolated integration database; runs repository/RLS integration tests and desktop checks.
Đặng Ngọc Minh Đức	Supervisor	Reviews the test plan and evidence; takes part in acceptance testing at Review 2 and the Mock Council.

3.2 Test Environment

The software, hardware and infrastructure used for testing are listed below.

Purpose	Tool	Provider	Version
Developer workstations	Windows 11 + Node.js + pnpm	Microsoft / OpenJS / pnpm	Node.js 24, pnpm 11.19.0
Local database for development	Docker Compose postgres service (pgvector image)	pgvector / PostgreSQL	pgvector/ pgvector:pg17
Isolated integration database in CI	Disposable PostgreSQL + pgvector	GitHub Actions / PostgreSQL	PostgreSQL 17; CI #268
Hosted development slots	Coolify on Ubuntu 24.04 with Supabase PostgreSQL, Auth and Storage	Coolify / Supabase	Shared dev server
Continuous integration runners	GitHub Actions hosted runners	GitHub	ubuntu-latest, windows-latest
Web client under test	Chromium-based browser, Firefox	Google / Mozilla	Current stable
Desktop client under test	Electron app on Windows	OpenJS Foundation	Electron 44.4.3
Email delivery for system tests	SMTP test mailbox	Configurable SMTP provider	-

3.3 Test Milestones

Test milestones follow the SEP490 15-week schedule (Review 2 in W8, system testing in W12, Faculty Council in W13).

Milestone Task	Start Date	End Date
Unit and API tests for Mainflow 1 and Mainflow 2 (W5-W8)	28/09/2026	25/10/2026
Migration and RLS integration rehearsal on PostgreSQL (W8)	19/10/2026	25/10/2026
Archive passing disposable PostgreSQL, RLS and release-	26/10/2026	01/11/2026

protocol CI evidence (W9)		
Unit and integration tests for remaining mainflows: search, AI retrieval, handover (W9-W11)	26/10/2026	15/11/2026
System testing, regression and defect fixing on the hosted slot (W12)	16/11/2026	22/11/2026
Acceptance testing with supervisor and Mock Council (W12-W13)	16/11/2026	29/11/2026
Final regression and test report update (W14)	30/11/2026	06/12/2026

4. Test Cases

Reported checkout evidence and test-case inventory are distinct. On feat/workspace-shell-polish, a Grok-supplied local checkout summary lists 387 tests: 380 passed, seven skipped and zero failed. Of those, UI contributes 168 passed, API 136 passed and seven skipped, Desktop seven, Domain five, Remote workflow 16, and Release/pull-request checks 48. Of those 209 documented entries, 92 unit conditions and 40 API route tests are Passed because their assertions ran in this checkout. Twelve unit conditions stay Not mapped: U11-04 has no unknown-status assertion, and U12-01 through U14-06 are Arden-AI pytest cases that were not part of this run. The seven live integration tests stay Skipped. The 50 manual system scenarios and eight acceptance demonstrations were not executed. The repository source inventory of 513 declarations is historical to its pinned revisions.

4.1 Unit Test Cases

The Unit Test workbook defines 104 concrete conditions across 19 function sheets, each with source-backed preconditions, inputs, expected confirmations and normal/abnormal/boundary classifications. A row is Passed when its assertion is in a file that passed on this checkout. U11-04 stays Not mapped because invitationStatus.test.ts does not assert an unknown status. U12-01 through U14-06 stay Not mapped because tests/test_answering.py, tests/test_chunking.py and tests/test_contracts.py were not in this Arden run.

Module	Test file	Cases	Result
canReadPublication	apps/api/src/knowledge/policy.test.ts	8	Passed
canReviewSnapshot	apps/api/src/knowledge/policy.test.ts	7	Passed
canSubmitSnapshot	apps/api/src/knowledge/policy.test.ts	7	Passed
planReviewDecision	apps/api/src/knowledge/review-decision.test.ts	7	Passed
authorizedReviewQueue	apps/api/src/knowledge/review-queue.test.ts	8	Passed
IndexingWorker.runOnce	apps/api/src/indexing/worker.test.ts	6	Passed
LocalPrivateContentStorage	apps/api/src/private-notes/content-storage.test.ts	6	Passed
apiRequestPolicy	apps/desktop/src/main/apiRequestPolicy.test.ts	7	Passed
SemanticDocument v1 parser	packages/domain/src/semantic-document.test.ts	7	Passed
notePreviewReducer	packages/ui/tests/noteDraftModel.test.ts	8	Passed
invitationState / visibleInvitations	packages/ui/tests/invitationStatus.test.ts	5	Passed
PermissionAwareAnswerer.answer	tests/test_answering.py	3	Not mapped
chunk_markdown	tests/test_chunking.py	2	Not mapped

IndexGenerationRequest	tests/test_contracts.py	6	Not mapped
citationPageForSource	apps/api/src/answers/citation-page.test.ts	3	Passed
knowledgeAudience selection	packages/ui/tests/knowledgeAudience.test.ts	3	Passed
LocalFlow1TestAuthProvider	apps/api/src/auth/local-flow1-test.test.ts	6	Passed
PDF viewer identity	packages/ui/tests/pdfDocument.test.ts	2	Passed
IndexGenerationManifest validator	apps/api/src/indexing/contracts.test.ts	3	Passed

Unit test case list:

ID	Test case	Module	Result
U01-01	ACTIVE group member; expected true: readable	UT-001	Passed
U01-02	No group membership; expected false: denied	UT-001	Passed
U01-03	REVOKED group membership; expected false: denied	UT-001	Passed
U01-04	SUSPENDED actor; expected false: denied	UT-001	Passed
U01-05	CURRENT pointer; legacy dates inverted; expected true: current pointer wins	UT-001	Passed
U01-06	ORGANIZATION_ALL audience; org matches; expected true: readable	UT-001	Passed
U01-07	DEPARTMENT_ALL audience; wrong department; expected false: denied	UT-001	Passed
U01-08	Reviewer role without reader scope; expected false: denied	UT-001	Passed
U02-01	Ordinary reviewer with live matching grant; expected true: authorized	UT-002	Passed
U02-02	Author reviews own ordinary submission; expected false: denied	UT-002	Passed
U02-03	Restricted class; no specialist level; expected false: denied	UT-002	Passed
U02-04	Org Admin self-review, no grant; expected true: admin exception	UT-002	Passed
U02-05	Dept reviewer on own department snapshot; expected true: authorized	UT-002	Passed
U02-06	Expired reviewer grant; expected false: denied	UT-002	Passed
U02-07	Org Admin final authority; expected true: authorized	UT-002	Passed
U03-01	Immutable snapshot and enforceable audience; expected true: can submit	UT-003	Passed
U03-02	Audience is unenforceable; expected false: denied	UT-003	Passed
U03-03	Responsible department = author department;	UT-003	Passed

	expected true: can submit		
U03-04	Different department chosen; expected false: denied	UT-003	Passed
U03-05	Disjoint scopes in submission; expected true: correct scope	UT-003	Passed
U03-06	Breadth not explicitly LIMITED; expected false: denied	UT-003	Passed
U03-07	Legacy organization-wide submission; expected legacy transition allowed	UT-003	Passed
U04-01	Final authorized one-step APPROVE; expected publication preparation intent	UT-004	Passed
U04-02	Wrong snapshot version; expected DENIED: STALE_SNAPSHOT	UT-004	Passed
U04-03	Content hash changed; expected DENIED: STALE_SNAPSHOT	UT-004	Passed
U04-04	Inactive or mismatched grant; expected DENIED: NOT_AUTHORIZED	UT-004	Passed
U04-05	Unenforceable audience; expected DENIED	UT-004	Passed
U04-06	REJECT with reason; expected no publication intent	UT-004	Passed
U04-07	Org Admin self-approves; expected authorized admin exception	UT-004	Passed
U05-01	Live matching reviewer; expected only authorized pending rows	UT-005	Passed
U05-02	Invalid-audience submission; expected remains visible to reject	UT-005	Passed
U05-03	Org Admin own pending submission; expected visible without grant	UT-005	Passed
U05-04	Live matching classification/team grant; expected visible	UT-005	Passed
U05-05	Restricted without specialist; expected hidden	UT-005	Passed
U05-06	Org-wide submission; department reviewer; expected hidden	UT-005	Passed
U05-07	Inactive reviewer or malformed policy data; expected fail closed	UT-005	Passed
U05-08	Duplicate joins and newer submissions; expected deduped newest first	UT-005	Passed
U06-01	Lease 300s default; expected returns 300 seconds	UT-006	Passed
U06-02	Lease shorter than request margin; expected throws config error	UT-006	Passed
U06-03	Superseded generation on complete; expected nonretryable superseded failure	UT-006	Passed
U06-04	One ready semantic snapshot job; expected indexes; complete activates atomically	UT-006	Passed
U06-05	Integrity-checked PDF snapshot; expected base64 payload roundtrips	UT-006	Passed

U06-06	No ready job; expected idle without side effects	UT-006	Passed
U07-01	Store and read Unicode text; expected roundtrip; hash verified	UT-007	Passed
U07-02	Read with incorrect SHA-256; expected integrity check failure	UT-007	Passed
U07-03	Path traversal reference; expected invalid private URI	UT-007	Passed
U07-04	Delete opaque object, read again; expected read rejects after deletion	UT-007	Passed
U07-05	Store PDF bytes; readPdf; expected byte-identical output	UT-007	Passed
U07-06	Read PDF as text; expected explicit PDF/text rejection	UT-007	Passed
U08-01	Bounded audience-group edit; expected allowed with precondition	UT-008	Passed
U08-02	Shared draft with idempotency key; expected allowed	UT-008	Passed
U08-03	PDF upload on specific route; expected larger body allowed	UT-008	Passed
U08-04	Missing or misplaced idempotency key; expected blocked	UT-008	Passed
U08-05	Existing private note and session routes; expected allowed	UT-008	Passed
U08-06	Known author/review/admin route; expected allowed	UT-008	Passed
U08-07	Method/path/precondition confusion; expected blocked	UT-008	Passed
U09-01	Valid IDs, formatting and links; expected round-trip identical	UT-009	Passed
U09-02	Missing stable block ID; expected invalid document	UT-009	Passed
U09-03	Duplicate stable IDs; expected invalid document	UT-009	Passed
U09-04	New plain text input; expected normalizes into blocks	UT-009	Passed
U09-05	Raw plain text parsed as stored doc; expected INVALID_PREFIX	UT-009	Passed
U09-06	Unsafe link URL; expected invalid document	UT-009	Passed
U09-07	Unknown canonical property; expected invalid document	UT-009	Passed
U10-01	Edit then switch back; expected draft remains unsaved	UT-010	Passed
U10-02	Save selected note while second dirty; expected other draft retained	UT-010	Passed
U10-03	Revert edit to saved value; expected dirty=false	UT-010	Passed
U10-04	Create new note while prior draft dirty; expected prior draft preserved	UT-010	Passed
U10-05	Reset active session; expected drafts cleared	UT-010	Passed
U10-06	Select unknown note ID; expected same state; no	UT-010	Passed

	exception		
U10-07	Discard only selected draft; expected other drafts intact	UT-010	Passed
U10-08	Remove never-saved note; expected saved notes intact	UT-010	Passed
U11-01	PENDING expires tomorrow; expected PENDING	UT-011	Passed
U11-02	PENDING expired yesterday; expected EXPIRED	UT-011	Passed
U11-03	Server ACCEPTED or REVOKED; expected server state retained	UT-011	Passed
U11-04	Unknown status; expected maps to REVOKED	UT-011	Not mapped
U11-05	Mixed list with duplicate joins; expected each counted once; pending first	UT-011	Passed
U12-01	Retrieved allowed chunk + cited generator; expected ANSWERED; citation grounded	UT-012	Not mapped
U12-02	Retrieved chunk from other generation; expected throws outside allow-list	UT-012	Not mapped
U12-03	Retriever returns zero chunks; expected INSUFFICIENT_EVIDENCE; no citation	UT-012	Not mapped
U13-01	Chunk text with overlap below size; expected offsets and parent IDs preserved	UT-013	Not mapped
U13-02	Overlap >= chunk size; expected error: overlap must be smaller	UT-013	Not mapped
U14-01	Valid exact camelCase payload; expected parsed UUID and content	UT-014	Not mapped
U14-02	Unexpected callbackUrl property; expected validation error	UT-014	Not mapped
U14-03	Future contractVersion; expected validation error	UT-014	Not mapped
U14-04	Snapshot text/hash disagree; expected validation error	UT-014	Not mapped
U14-05	Raw PDF bytes match hash; expected pdf_bytes matches input	UT-014	Not mapped
U14-06	Export generated schema; expected concrete contract objects	UT-014	Not mapped
U15-01	PDF block with page number; expected source PDF page returned	UT-015	Passed
U15-02	Page stored only on parent block; expected parent page returned	UT-015	Passed
U15-03	Non-PDF block; expected no PDF navigation page	UT-015	Passed
U16-01	Organization and department scopes; expected only matching managed options	UT-016	Passed
U16-02	Switch ORGANIZATION to DEPARTMENT; expected ALL_MEMBERS removed	UT-016	Passed

U16-03	Empty/unknown/duplicate selected audiences; expected selection rejected	UT-016	Passed
U17-01	Known loopback/local DB; expected configuration accepted	UT-017	Passed
U17-02	External DB binding; expected configuration rejected	UT-017	Passed
U17-03	Synthetic example.test identities; expected accounts accepted	UT-017	Passed
U17-04	Unknown account sign-in; expected INVALID_CREDENTIALS	UT-017	Passed
U17-05	Revoke in-memory session; expected resolves unauthenticated	UT-017	Passed
U17-06	Runtime non-test database; expected rejects local-only target	UT-017	Passed
U18-01	Citation page changes; expected viewer identity changes	UT-018	Passed
U18-02	First page and unsafe filename; expected stable first-page identity; safe name	UT-018	Passed
U19-01	Text and PDF requests; expected MIME explicit; identity preserved	UT-019	Passed
U19-02	Strict v1 manifest and valid digest; expected manifest accepted	UT-019	Passed
U19-03	Identity drift or wrong chunk digest; expected manifest rejected	UT-019	Passed

4.2 Integration Test Cases

The pinned source inventory lists 40 Fastify API route declarations and seven live auth/database integration declarations. The current checkout report explicitly identifies the seven integration entries below as Skipped (not Passed); those seven are the sole reported skips within the API area. The other 40 declarations are the tests in apps/api/src/app.test.ts. That file passed, so those rows are labeled Passed. One skip requires all four Supabase test-auth variables: ARDEN_TEST_SUPABASE_URL, ARDEN_TEST_SUPABASE_PUBLISHABLE_KEY, ARDEN_TEST_SUPABASE_EMAIL and ARDEN_TEST_SUPABASE_PASSWORD. Six database skips require ARDEN_TEST_DATABASE_URL, pointing to a superuser connection to an isolated database named arden_test_*. Neither integration environment was configured for the reported local checkout.

ID	Test case	Module	Result
IT-API-001	keeps health routes public and reports readiness failures safely	API routes	Passed
IT-API-002	answers only from the caller	API routes	Passed
IT-API-003	drops the answer when publication access changes before citations are returned	API routes	Passed
IT-API-004	signs in through the Supabase adapter and returns only the caller	API routes	Passed
IT-API-005	completes Google PKCE sign-in and requires a one-time Arden password	API routes	Passed
IT-API-006	handles sign-up validation and duplicate account	API routes	Passed

	rejection		
IT-API-007	handles forgot password requests, verification proxy, and password reset	API routes	Passed
IT-API-008	allows the explicitly injected Kiet Local test actor to use core routes without a login cookie	API routes	Passed
IT-API-009	creates an organization with the verified caller as its first admin and primary-department member	API routes	Passed
IT-API-010	exposes organization administration and audience/department mutations only to the Org Admin	API routes	Passed
IT-API-011	requires a signed-in verified email and validates organization setup input	API routes	Passed
IT-API-012	keeps instance administration separate from organization administration	API routes	Passed
IT-API-013	does not allow revoking the last active Arden System Admin	API routes	Passed
IT-API-014	sends opaque invitations and accepts them only for the matching authenticated email once	API routes	Passed
IT-API-015	does not acknowledge invitation delivery when SMTP reports a failure	API routes	Passed
IT-API-016	labels local-test invitations as saved without sending email	API routes	Passed
IT-API-017	revokes a pending invitation and blocks later acceptance	API routes	Passed
IT-API-018	rejects bad credentials, missing sessions, and malformed sign-in payloads	API routes	Passed
IT-API-019	keeps invitation account setup non-enumerating and never starts an authenticated session	API routes	Passed
IT-API-020	signs out through Supabase and clears the session cookies	API routes	Passed
IT-API-021	creates and reads only the authorized owner	API routes	Passed
IT-API-022	returns the same safe 404 for forged owner IDs and cross-organization guesses	API routes	Passed
IT-API-023	checks role permission before owner scope and never grants managers another member	API routes	Passed
IT-API-024	rejects missing and stale update versions without acknowledging an overwrite	API routes	Passed
IT-API-025	updates with an expected version and advances the immutable version number	API routes	Passed
IT-API-026	validates tenant identifiers, unknown fields, NULs, and oversized request bodies	API routes	Passed
IT-API-027	creates an independent shared copy of the author	API routes	Passed
IT-API-028	denies missing sessions, foreign organizations or versions, and malformed create requests	API routes	Passed
IT-API-029	keeps a content copy when database commit	API routes	Passed

	outcome is unknown		
IT-API-030	creates a server-backed new shared draft, then lists and reopens only that actor	API routes	Passed
IT-API-031	accepts new drafts and reviewer grants without a classification (decision 0013)	API routes	Passed
IT-API-032	creates whole-organization grants and rejects mismatched scope metadata (decision 0015)	API routes	Passed
IT-API-033	preserves an uploaded PDF through metadata, submission, review and preparation	API routes	Passed
IT-API-034	rejects malformed, oversized and cross-organization PDF uploads	API routes	Passed
IT-API-035	exposes active knowledge metadata options only within the actor	API routes	Passed
IT-API-036	rejects invalid new shared drafts and does not reveal another organization	API routes	Passed
IT-API-037	updates a shared draft as a new version only when If-Match is current	API routes	Passed
IT-API-038	routes submit, scoped review and preparation without exposing the private source	API routes	Passed
IT-API-039	keeps the working draft mutable while the submitted snapshot remains pending	API routes	Passed
IT-API-040	routes desktop-compatible sign-in through the same HttpOnly Supabase session	API routes	Passed
IT-DB-001	authenticates the synthetic user and rejects the session after sign-out.	Auth integration	Skipped
IT-DB-002	uses a non-superuser worker login without API private-content access.	DB repository	Skipped
IT-DB-003	enforces owner-only drafts, member creation, conflicts, and audit records through RLS.	DB repository	Skipped
IT-DB-004	copies only an authorized private version and keeps shared rows invisible to another member.	DB repository	Skipped
IT-DB-005	shows invited members to their Org Admin and permits a reviewer grant without exposing profiles to other actors.	DB repository	Skipped
IT-DB-006	keeps organization and department reviewer scopes disjoint and exposes grant availability.	DB repository	Skipped
IT-DB-007	durably enqueues and atomically activates a different member's approved snapshot.	DB repository	Skipped

4.3 System Test Cases

The 50 designed system scenarios cover identity (7), organization workflows (6), private content (5), governance (9), indexing (6), Ask Arden (6), administration (4) and release safety (7). They are manual/functional documentation and have no recorded tester, date, environment or observed results here; the 387 automated-test summary cannot mark them Passed. All retain Not executed status pending evidence.

ID	Scenario	Expected result	Status

TC-05-001	[AUTH] Email account signup. An unauthenticated user submits a valid new email, password and account details.	The sign-up API validates the request, rejects malformed input and follows its configured verification flow.	Not executed
TC-05-002	[AUTH] Duplicate signup. Submit the signup form with an email already registered.	The duplicate is rejected without creating a second account.	Not executed
TC-05-003	[AUTH] Password reset. Request a reset then complete the verification and password-change flow.	The reset uses the verified caller flow; invalid or expired credentials cannot change another account.	Not executed
TC-05-004	[AUTH] Google PKCE callback. Begin Google sign-in, return to the Arden callback and finish the configured one-time password step.	PKCE exchange remains bound to the correct callback and caller; a required setup step cannot be bypassed.	Not executed
TC-05-005	[AUTH] Signed-out session invalidation. Sign in then sign out, then reuse the prior session.	The server rejects the stale session and does not return an authenticated caller.	Not executed
TC-05-006	[AUTH] Desktop API allowlist. Attempt a desktop bridge request with an unsupported method or endpoint.	The main-process request policy rejects the request before forwarding.	Not executed
TC-05-007	[AUTH] Health and readiness privacy. Request public health/readiness endpoints during an underlying dependency failure.	Public health remains accessible; readiness reports failure without exposing secrets.	Not executed
TC-05-008	[ORG] Create organization. An authenticated account creates a new organization.	The creator becomes its first organization admin and primary-department member.	Not executed
TC-05-009	[ORG] Invite existing and new members. An admin invites a new email and tries inviting an already-member email.	New invitation is created without membership mutation; existing member email is rejected case-insensitively.	Not executed
TC-05-010	[ORG] Invitation token and account binding. Open an invitation using the wrong signed-in account.	The wrong account cannot accept or gain membership.	Not executed
TC-05-011	[ORG] Invitation expiry and renewal. Attempt acceptance at the exact expiry boundary, then renew an eligible expired invitation.	Expired invitation is denied and renewal issues a new valid seven-day invitation.	Not executed
TC-05-012	[ORG] Membership assignment waiting room. Sign in as an active member with no department assignment.	The member is placed in the waiting state; unrelated private-workspace access is not assumed.	Not executed
TC-05-013	[ORG] Department scope separation. An organization admin assigns departments and reviewer grants, then inspect as an unrelated actor.	Visibility follows organization and department boundaries; unrelated profiles are not exposed.	Not executed
TC-05-014	[PRIVATE] Create and edit own note. An authorized user creates and saves a private note and later edits it.	The note remains private to its owner, versioned and retrievable through authorized access.	Not executed
TC-05-015	[PRIVATE] Cross-user private note denial. An unrelated member attempts to read/copy/edit another member's private note.	Row-level authorization denies access and does not leak content.	Not executed
TC-05-016	[PRIVATE] Concurrent edit conflict. Use an out-of-date note version while another save	The conflict is surfaced and unsaved editor changes remain	Not executed

	has already succeeded.	recoverable.	
TC-05-017	[PRIVATE] Unsaved note switching. Edit a note without saving, switch to another note and back.	Unsaved data is preserved in the client draft model until saved or explicitly discarded.	Not executed
TC-05-018	[PRIVATE] Opaque PDF content storage. Upload PDF note content and attempt a client-controlled storage path.	Bytes are stored with opaque references and hashes; unsafe client-supplied paths are rejected.	Not executed
TC-05-019	[GOVERNANCE] Private-to-shared copy. Create a shared draft from owned private content; retry using a private note from another actor.	Authorized copy succeeds; inaccessible content cannot become shared.	Not executed
TC-05-020	[GOVERNANCE] Responsible-department ownership. An author chooses a responsible department different from their primary department.	Submission policy rejects an unsupported department override.	Not executed
TC-05-021	[GOVERNANCE] Immutable review submission. Submit a draft for review and edit its working copy afterwards.	The pending submission retains its original immutable snapshot.	Not executed
TC-05-022	[GOVERNANCE] Reviewer grant and self-review. Attempt approval as the author, as an unrelated member, then as a valid matching reviewer.	Only a currently eligible reviewer can approve; ordinary self-review is blocked.	Not executed
TC-05-023	[GOVERNANCE] Restricted classification reviewer. Attempt Restricted-material review with a reviewer without specialist/senior grant.	Review is denied unless the classification-specific grant criteria are met.	Not executed
TC-05-024	[GOVERNANCE] Invalid publication audience. Submit a group/department audience outside permitted scope.	Out-of-scope audience is rejected; invalid pending submissions remain rejectable.	Not executed
TC-05-025	[GOVERNANCE] Approval is not immediate publication. Approve a valid review while indexing is incomplete.	Approval prepares publication; readers cannot see a new active snapshot until index-ready activation.	Not executed
TC-05-026	[GOVERNANCE] Old publication retained on revision. Revise an already published document and submit a new snapshot.	Existing active version remains readable until the new approved indexed generation activates.	Not executed
TC-05-027	[GOVERNANCE] Publication validity period. Read a publication before its effective date and after its expiry.	Client labels upcoming/expired states and access policy respects active validity.	Not executed
TC-05-028	[INDEX] Indexed snapshot identity. Submit a valid content snapshot for indexing and inspect resulting generation manifest.	Content hashes and source identity match the immutable input; manifest is strict and digest verified.	Not executed
TC-05-029	[INDEX] Superseded generation retry. Supersede a pending index job before retrying the old generation.	Worker does not publish or re-index a superseded generation.	Not executed
TC-05-030	[INDEX] PDF index payload integrity. Index a PDF snapshot through the Arden/Arden-AI boundary.	PDF is sent with MIME type and integrity checks without altering snapshot identity.	Not executed
TC-05-031	[INDEX] AI index endpoint service authentication. Call the AI index generation	The endpoint rejects unauthenticated calls.	Not executed

	endpoint without the configured service credential.		
TC-05-032	[INDEX] Deterministic document chunking. Compile a structured document twice with the same content and chunk settings.	Chunk boundaries, offsets and parent mapping are stable.	Not executed
TC-05-033	[INDEX] Image-only PDF handling. Submit a PDF with no extractable text to the AI compiler.	The PDF compiler rejects unsupported image-only content rather than inventing text.	Not executed
TC-05-034	[AI] Answer service authentication. Call the AI answer endpoint without valid inter-service authentication.	The endpoint rejects the request.	Not executed
TC-05-035	[AI] Retrieval allowlist. Ask a question with an accessible publication set and inject other generation IDs into retrieval.	Only core-provided authorized generation IDs may contribute evidence.	Not executed
TC-05-036	[AI] Answer citation evidence. Ask a question whose retrieved evidence contains citations.	Answer cites only retrieved evidence; unrelated sources are not returned.	Not executed
TC-05-037	[AI] No-evidence abstention. Ask a question with no eligible indexed source.	AI returns an honest no-evidence response with no fabricated citations.	Not executed
TC-05-038	[AI] Citation-time permission recheck. Revoke the caller's publication access after retrieval but before the response is returned.	The core API drops the answer and citations instead of leaking revoked content.	Not executed
TC-05-039	[AI] PDF citation page navigation. Select a citation backed by a specific page in a PDF.	The UI navigates to the correct page with safe viewer identity and filenames.	Not executed
TC-05-040	[ADMIN] Organization member privacy. List and manage members as an organization admin and then as an unrelated role.	Organization admins see authorized member administration; unrelated actors cannot inspect profiles.	Not executed
TC-05-041	[ADMIN] Reviewer scope filtering. View pending review queue with department-scoped and organization-scoped reviewer grants.	Only currently assigned and properly scoped submissions appear.	Not executed
TC-05-042	[ADMIN] Invitation mailer config safety. Start with incomplete SMTP settings and attempt sending an invitation.	Incomplete configuration fails explicitly without exposing credentials.	Not executed
TC-05-043	[ADMIN] Audit trail on protected writes. Perform an authorized private/shared write and inspect audit behavior.	Authorized storage persists the relevant audit record; unauthorized cross-user write is blocked.	Not executed
TC-05-044	[RELEASE] Migration checksum drift. Modify SQL bytes of a previously reviewed migration and run inventory verification.	Checksum and immutable-history checks reject the change.	Not executed
TC-05-045	[RELEASE] Migration plan operator fence. Provide a pending operator-class migration to the deployment planner.	Planner blocks unattended application before destructive/incompatible changes.	Not executed
TC-05-046	[RELEASE] Disposable DB migration repeat. Apply migration chain twice to disposable PostgreSQL.	Second full run is an idempotent no-op and schema authorization tests pass.	Not executed
TC-05-047	[RELEASE] Release identity binding. Alter repository, branch, environment or source	Release receiver rejects the mismatched identity before	Not executed

	identity of a signed release request.	mutation.	
TC-05-048	[RELEASE] Tampered image upload. Modify uploaded release payload after checksums were established.	Receiver rejects tampered upload without activating new images.	Not executed
TC-05-049	[RELEASE] Release rollback and retention. Simulate an interrupted activation and inspect rollback and cleanup.	Previous verified release can be restored; in-use/previous-environment images are retained.	Not executed
TC-05-050	[RELEASE] Exclusive remote writer. Start competing shared-development backend sync sessions.	Only one writer receives the lease; a rejected competitor cannot release the valid writer's lock.	Not executed

4.4 Acceptance Test Cases

Eight acceptance demonstrations consolidate key end-user journeys for supervisor review. These remain Not signed off pending a dated supervised run and artifacts; the automated checkout does not substitute for user-acceptance evidence.

ID	Scenario	Expected result	Status
AT-01	Demo an end-user registration, session and signed-out access path.	An identity cannot reuse a revoked session; owner context remains isolated.	Not signed off
AT-02	Invite an account, accept an organization invitation, assign department and role.	Token binding, expiry, membership and department restrictions match policy.	Not signed off
AT-03	Create, edit and reopen a private note and stored PDF as two different users.	Only the owner can read or update; conflicting saves preserve the correct version.	Not signed off
AT-04	Copy a draft, submit immutable review snapshot and approve as an authorized reviewer.	Review uses live grants and exact snapshot; unauthorized or stale decisions are denied.	Not signed off
AT-05	Index approved content, then supersede a generation during retry.	Only the authorized current generation activates with matching digest and PDF identity.	Not signed off
AT-06	Ask Arden with allowed, forbidden and missing evidence.	Answers cite allowed text; forbidden source is absent and no-evidence queries abstain.	Not signed off
AT-07	Show administration, invitation mailer configuration and protected-action audit.	Sensitive fields remain private; role boundaries and audit facts are enforced.	Not signed off
AT-08	Rehearse image verification, migration and release operator protocol in a disposable slot.	Wrong identity/digest is rejected; migration and rollback receipts are archived.	Not signed off

5. Test Reports

Evidence as of 31/10/2026. Newest reported local checkout: feat/workspace-shell-polish, 387 tests collected, 380 passed, seven skipped, zero failed (Local run on commit 045a7b5 of pnpm test, pnpm test:remote, and the release and pull-request node tests). The seven skips are one Supabase live-auth check and six PostgreSQL row-level-security checks. Separately, older GitHub Actions #268 at pinned Arden 953be54, Arden-AI cc0b4d2, Arden-Knowledge d4020b4 reported successful Ubuntu, Windows, PostgreSQL and image/security jobs, and a skipped deployment job. That skipped deployment job is not one of the seven API integration skips.

Reported checkout results (separate from documented test-case inventory)

Area / suite	Total	Passed	Failed / skipped / status
UI	168	168	0 failed / 0 skipped
API	143	136	0 failed / 7 skipped
Desktop	7	7	0 failed / 0 skipped
Domain	5	5	0 failed / 0 skipped
Remote workflow	16	16	0 failed / 0 skipped
Release / pull-request checks	48	48	0 failed / 0 skipped
Reported checkout total	387	380	0 failed / 7 skipped
Curated Unit Test conditions	104	N/A	No per-case results
Manual system scenarios	50	N/A	Pending execution
Acceptance demonstrations	8	N/A	Pending sign-off

Defects found

ID	Description	Severity	Status
—	No failures were reported for the 387-test checkout. No new reproduced defect was supplied; earlier defect candidates require separate verification.	—	0 reported failures

Test analysis

Result. The reported checkout finished with 380 passed, seven skipped and zero failed among 387 collected tests (380/387 passed; 7/387 skipped). API accounts for all seven skips (136 passed, seven skipped); UI passed 168, Desktop passed seven, Domain five, Remote workflow 16, and Release/pull-request checks 48. These are reported automated-run figures, not independently reproduced by this documentation update. The 47-row integration catalog is 40 Passed route tests and seven Skipped live checks. Ninety-two curated unit conditions are Passed and twelve remain Not mapped. The 50 manual system scenarios and eight acceptance demonstrations were not executed.

Coverage of requirements. The reported checkout demonstrates automated execution in six areas: UI, API, Desktop, Domain, Remote workflow and Release/pull-request checks; this is a test-count breakdown, not line/branch or requirements-coverage percentage. The older source inventory identifies 513 declarations across 75 modules at pinned revisions and supports traceability for authorization, private ownership, reviewer scope, publishing, indexing, AI retrieval and citations, desktop policies and release safety. No proof has been supplied that all 513 source declarations were run on the new branch; designed manual and acceptance coverage remains pending.

Risks and recommendations. (1) Attach a saved Vitest log for commit 045a7b5. The local commands were pnpm test, pnpm test:remote, and the release and pull-request node tests. (2) Configure a nonproduction Supabase auth tenant and a disposable arden_test_* PostgreSQL superuser fixture, then rerun the seven explicitly skipped checks. (3) Never run repository integration fixture cleanup against production. (4) U11-04 and the eleven Arden-AI unit rows are still Not mapped; the 50 system scenarios and eight acceptance demonstrations still need a signed run. (5) Archive Arden-AI pytest

evidence separately; its inclusion in the 387 count was not established. (6) Reconcile the older 513 source declarations and CI #268 with the new branch at its exact SHA before computing coverage or announcing a release gate.